

GRADUATION PROJECT — AI ENGINEERING BOOTCAMP

The Student's Channel Brain

Final 38-Task Evaluation Report

MIT 18.06 Linear Algebra Tutor — Final frozen system evaluation across all 34 lectures

38 Tasks	32 Passed	84.2% Overall pass rate	92.9% Timestamp / citation accuracy*
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*26 correct out of 28 tasks where the Excel field "Timestamp Correct?" was explicitly assessed. The dedicated Source/Timestamp Check category passed 2/2.

1. Evaluation Design

Purpose. This evaluation measures the final end-to-end behavior of the deployed lecture tutor after the retrieval configuration was frozen. The 38 tasks were executed manually in the final web application and judged against predefined pass criteria. LangSmith captured technical traces where available. No RAG tuning was performed during the final run.

Frozen configuration. Lectures 1–34; 1200-character chunks; vector retrieval k=8; CrossEncoder reranker ON; rerank top_n=3; GPT-5.6 Luna with reasoning_effort="high"

Coverage. The test set spans nine behavioral categories: direct concepts, lecture-specific questions, comparisons, multi-turn memory/follow-up, out-of-scope handling, quiz generation/grading, weak-topic tracking, course building, and source/timestamp validation.

2. Overall Results

32 of 38 tasks passed, for an overall end-to-end pass rate of 84.2%. The system was strongest on direct concept questions, out-of-scope handling, weak-topic tracking, and the dedicated source/timestamp checks (all 100%). The weakest categories were memory/follow-up (60.0%) and course builder behavior (66.7%).

Category	Tasks	Pass	Fail	Pass rate
Direct Concept	6	6	0	100.0%
Lecture-Specific	6	5	1	83.3%
Comparison	5	4	1	80.0%
Memory / Follow-up	5	3	2	60.0%
Out-of-Scope	4	4	0	100.0%
Quiz Gen/Grading	4	3	1	75.0%
Weak-Topic Tracking	3	3	0	100.0%
Course Builder	3	2	1	66.7%
Source/Timestamp Check	2	2	0	100.0%

3. Category-Level Interpretation

- **Direct Concept — 6/6.** Grounded definitions and explanations were consistently correct and accompanied by valid lecture evidence.
- **Lecture-Specific — 5/6.** Lecture filtering generally worked; the single failure was a content-completeness issue in Lecture 33 rather than cross-lecture leakage.
- **Comparison — 4/5.** Most multi-concept comparisons were correct, but one case selected supporting evidence from the wrong foundational lecture.

- **Memory / Follow-up — 3/5.** Three conversational-memory tasks passed. Both failures were runtime/syntax interruptions rather than demonstrated conceptual-memory errors.
- **Out-of-Scope — 4/4.** The tutor reliably declined unrelated or invalid-course requests instead of hallucinating course evidence.
- **Quiz Gen/Grading — 3/4.** Quiz generation and normal grading worked; malformed/ambiguous answer handling remains too permissive.
- **Weak-Topic Tracking — 3/3.** The system correctly represented no-topic, single-topic, and multiple-topic weak-state cases.
- **Course Builder — 2/3.** Explicit topic and lecture-scoped modules worked; automatic topic selection from session state failed once.
- **Source/Timestamp Check — 2/2.** Dedicated inspection confirmed the required source fields, lecture scoping, timestamp format, and timestamped video-link behavior.

4. Failure Analysis

Six failures were recorded. They are concentrated in completeness/evidence selection, runtime robustness, ambiguous quiz input handling, and session-state selection rather than a broad collapse of retrieval quality.

ID	Area	Failure type	Observed issue
12	Lecture-Specific	Scope/content completeness	Lecture 33 answer omitted the pseudoinverse, although the pass criteria required left inverse, right inverse, and pseudoinverse.
15	Comparison	Wrong lecture / evidence selection	Eigendecomposition vs. SVD comparison cited Lectures 22/23 instead of the required foundational Lecture 21 evidence.
21	Memory / Follow-up	Runtime / transient error	A transient syntax error caused the task to fail; resending succeeded and the content was correct.
22	Memory / Follow-up	Runtime / transient error	A SyntaxError interrupted the multi-turn determinant quiz / recall workflow.
30	Quiz Gen/Grading	Grounding / input handling	The grader silently interpreted an ambiguous answer instead of asking the student to answer with A-D.
36	Course Builder	State tracking	With no topic supplied, the course builder fell back to Lecture 31 instead of the first recorded weak topic, Lecture 21.

5. Timestamp and Citation Evaluation

Of 28 tasks explicitly assessed with the “Timestamp Correct?” field, 26 were marked Y and 2 were marked N, giving a 92.9% assessed timestamp/citation correctness rate.

Measure	Result
Timestamp/citation assessed	28

Measure	Result
Correct (Y)	26
Incorrect (N)	2
Not marked / N/A	10
Dedicated Source/Timestamp tasks passed	2/2 (100%)

Interpretation caution. The two N judgments correspond to Task 12 (required pseudoinverse content omitted) and Task 15 (wrong lecture evidence). They should therefore be treated as citation/grounding failures in the assessed task, not as proof of a “correct answer but wrong timestamp within the correct lecture” failure. The final run did not document that specific failure pattern. The two dedicated source/timestamp inspection tasks both passed.

6. Conclusions and Limitations

- The final system achieved 84.2% task-level success across a deliberately mixed 38-task end-to-end test set covering all major product behaviors.
- Grounded single-turn tutoring, scope control, weak-topic tracking, and dedicated citation/timestamp behavior were reliable in this run.
- The principal remaining weaknesses are runtime robustness in multi-turn flows, evidence selection for comparison questions, validation of ambiguous quiz answers, and automatic course-builder topic selection from session state.
- The 38-task set is large enough to satisfy the project requirement but remains a small sample relative to the full 34-lecture course; category percentages can move materially with one or two cases.
- Manual Pass/Fail and timestamp judgments provide transparent product-level evaluation, but they are not a substitute for larger repeated or automated evaluations.

Recommended next improvements

- Add strict input validation before quiz grading.
- Make the course-builder weak-topic selection deterministic.
- Harden multi-turn UI/runtime handling.
- Use query decomposition or lecture-aware evidence constraints for comparisons.
- Expand automated citation validation over more lectures.

Evidence Basis

- Notebook 06: final frozen configuration, task counts, category summary, overall 32/38 result, and original six-failure analysis.
- Lecture_Tutor_38_Task_Final_Results.xlsx: final row-level judgments, notes, LangSmith Run IDs where recorded, and the “Timestamp Correct?” field used for the citation/timestamp metric.

- Lecture_Tutor_38_Task_Evaluation_Set.xlsx: task definitions, expected tools, and pass criteria used to judge each result.